MAULANA AZAD NATIONAL INATITUTE OF TECHNOLOGY,
BHOPAL - 462003

Name of the Program:	B.Tech.	Semester	III	Year	**2nd**
Name of the Course:	EMEC-I				
Course Code:	EE-212				
Core/Elective/Other:	Core				
Pre-requisites:	Fundamentals of Basic Electrical Engineering				

Course Outcomes:	
1	Student has to evaluate the problems of 3-phase systems with balanced and unbalanced load.
2	Student has to evaluate and analyses the performance of DC Machines and to apply concept of speed control in DC Motor.
3	Student has to apply concept of 3-phase transformer for the different applications
4	Students has to evaluate and analyses the performance of transformer and DC machines using indirect testing
Description of Content in Brief:	
1.	**Polyphase circuits:** Three-phase systems with balanced & unbalanced load, measurement of power and power factor in three phase circuits
2.	**DC Generators:** Principle of operation, construction, lap and wave windings, emf equation, types of DC generators, voltage buildup, critical field resistance and critical speed, Armature reaction and commutation, methods of improving commutation, Characteristics, parallel operation
3.	**DC Motors:** Principle of operation, back EMF, torque equation, types of DC motors, characteristics, methods of speed control, types of starters, Losses and efficiency
4.	**Polyphase Transformers:** Review of single phase transformer,Three phase transformer: Principle of operation, vector groups, open delta connection, Scott connection, Auto transformers, harmonic reduction in phase voltages.
5.	**Testing of transformers and DC machines:** Testing of DC machines: Swinburne's test, Hopkinson's test, OC/SC and Sumpner's test on transformer
List of Text Books:	
1	P.S.Bhimbra, Electrical Machine, Khanna, 1 January 2011
2	Nagrath& Kothari, Electrical Machines, 4 edition, McGraw Hill Education, 7 July 2010
List of Reference Books	
1	M. G. Say, Performance & design of A.C. Machines, 3rd edition , CBS, 1 December 2005
2	Fitzerald Kingsley Otmans, Electrical Machines, 7 edition , McGraw-Hill Education, 1 March 2013
3	Charles. I. Hubert, Electric Machine, 2 edition, Prentice Hall, 16 October 2001
4	J.R. Cogdell, Foundation of Electric Power, Pck edition , Addison Wesley Longman, May 1, 2003
5	M. G. Say, Performance & design of A.C. Machines, 3rd edition , CBS, 1 December 2005
URLs:	
1	https://nptel.ac.in/courses/108/105/108105017/